Packet Tracer - Use Ping and Traceroute to Test Network Connectivity

Addressing Table

Device	Interface	IP Address / Prefix		Default Gateway
R1	G0/0	2001:db8:1:1::1/64		N/A
	G0/1	10.10.1.97	255.255.255.224	N/A
	S0/0/1	10.10.1.6	255.255.255.252	N/A
		2001:db8:1:2::2/64		
		fe80::1		
R2	S0/0/0	10.10.1.5	255.255.255.252	N/A
		2001:db8:1:2::1/64		
	S0/0/1	10.10.1.9	255.255.255.252	N/A
		2001:db8:1:3::1/64		
		fe80::2		
R3	G0/0	2001:db8:1:4::1/64		N/A
	G0/1	10.10.1.17	255.255.255.240	N/A
	S0/0/1	10.10.1.10	255.255.255.252	N/A
		2001:db8:1:3::2/64		
		fe80::3		
PC1	NIC	*10.10.1.98*	*255.255.255.224*	*10.10.1.97*
PC2	NIC	*2001:DB8:1:1::2/64*		*FE80::1*
PC3	NIC	*10.10.1.18*	*255.255.255.240*	*10.10.1.17*
PC4	NIC	*2001:DB8:1:4::2/64*		*FE80::3*

Objectives

Part 1: Test and Restore IPv4 Connectivity

Part 2: Test and Restore IPv6 Connectivity

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Scenario

There are connectivity issues in this activity. In addition to gathering and documenting information about the network, you will locate the problems and implement acceptable solutions to restore connectivity.

Note: The user EXEC password is **cisco**. The privileged EXEC password is **class**.

Instructions

Part 1: Test and Restore IPv4 Connectivity

Step 1: Use ipconfig and ping to verify connectivity.

a. Click **PC1** and open the **Command Prompt**.

b. Enter the **ipconfig /all** command to collect the IPv4 information. Complete the **Addressing Table** with the IPv4 address, subnet mask, and default gateway.

c. Click **PC3** and open the **Command Prompt**.

d. Enter the **ipconfig /all** command to collect the IPv4 information. Complete the **Addressing Table** with the IPv4 address, subnet mask, and default gateway.

e. Use the **ping** command to test connectivity between **PC1** and **PC3**. The ping should fail.

Step 2: Locate the source of connectivity failure.

a. From **PC1**, enter the necessary command to trace the route to **PC3**.

What is the last successful IPv4 address that was reached?

10.10.1.97 (R1's G0/1, PC1's default gateway).

b. The trace will eventually end after 30 attempts. Enter **Ctrl+C** to stop the trace before 30 attempts.

c. From **PC3**, enter the necessary command to trace the route to **PC1**.

What is the last successful IPv4 address that was reached?

10.10.1.17 (R3's G0/1, PC3's default gateway).

d. Enter **Ctrl+C** to stop the trace.

e. Click **R1**. Press **ENTER** and log in to the router.

f. Enter the **show ip interface brief** command to list the interfaces and their status. There are two IPv4 addresses on the router. One should have been recorded in Step 2a.

What is the other?

10.10.1.6 (R1's Serial0/0/1).

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g. Enter the **show ip route** command to list the networks to which the router is connected. Note that there are two networks connected to the **Serial0/0/1** interface.

What are they?

10.10.1.4/30 (connected, C) and 10.10.1.6/32 (local, L).

h. Repeat steps 2e through 2g with **R3** and record your answers.

R3: the other IPv4 address is 10.10.1.10 (Serial0/0/1). The two networks on Serial0/0/1 are 10.10.1.8/30 (connected, C) and 10.10.1.10/32 (local, L). The last successful hop from PC3 was 10.10.1.17.

i. Click **R2**. Press **ENTER** and log into the router.

j. Enter the **show ip interface brief** command and record your addresses.

R2: Serial0/0/0 = 10.10.1.2 (incorrect), Serial0/0/1 = 10.10.1.9, G0/0 and G0/1 unassigned and administratively down.

k. Run more tests if it helps visualize the problem. Simulation mode is available.

Step 3: Propose a solution to solve the problem.

Compare your answers in Step 2 to the documentation you have available for the network.

What is the error?

R2's Serial0/0/0 is configured with 10.10.1.2, but the addressing table and topology call for 10.10.1.5/30. 10.10.1.2 is not in the same subnet as R1's Serial0/0/1 (10.10.1.6/30), so R1 and R2 cannot communicate and R1 and R3 never learn routes to each other's LANs.

What solution would you propose to correct the problem?

Reconfigure R2's Serial0/0/0 with the correct address: interface serial 0/0/0, then ip address 10.10.1.5 255.255.255.252.

Step 4: Implement the plan.

Implement the solution you proposed in Step 3b.

Step 5: Verify that connectivity is restored.

a. From **PC1** test connectivity to **PC3**.

b. From **PC3** test connectivity to **PC1**.

Is the problem resolved?

Yes. Pings from PC1 to PC3 and from PC3 to PC1 now succeed.

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Step 6: Document the solution.

IPv4: R2's Serial0/0/0 was configured as 10.10.1.2 instead of 10.10.1.5/30, so R1 and R2 were on different subnets. I changed it to 10.10.1.5 255.255.255.252 on R2. PC1 and PC3 can now ping each other.

Part 2: Test and Restore IPv6 Connectivity

Step 1: Use ipv6config and ping to verify connectivity.

a. Click **PC2** and open the **Command Prompt**.

b. Enter the **ipv6config /all** command to collect the IPv6 information. Complete the **Addressing Table** with the IPv6 address, subnet prefix, and default gateway.

c. Click **PC4** and open the **Command Prompt**.

d. Enter the **ipv6config /all** command to collect the IPv6 information. Complete the **Addressing Table** with the IPv6 address, subnet prefix, and default gateway.

e. Test connectivity between **PC2** and **PC4**. The ping should fail.

Step 2: Locate the source of connectivity failure.

a. From **PC2**, enter the necessary command to trace the route to **PC4**.

What is the last successful IPv6 address that was reached?

2001:DB8:1:3::2 (R3's Serial0/0/1). The trace passed 2001:DB8:1:1::1 (R1) and 2001:DB8:1:2::1 (R2) before timing out.

b. The trace will eventually end after 30 attempts. Enter **Ctrl+C** to stop the trace before 30 attempts.

c. From **PC4**, enter the necessary command to trace the route to **PC2**.

What is the last successful IPv6 address that was reached?

None. Every hop timed out because PC4 could not reach its default gateway.

d. Enter **Ctrl+C** to stop the trace.

e. Click **R3**. Press **ENTER** and log in to the router.

f. Enter the **show ipv6 interface brief** command to list the interfaces and their status. There are two IPv6 addresses on the router. One should match the gateway address recorded in Step 1d.

Is there a discrepancy?

Yes. PC4's default gateway is FE80::2, but R3's G0/0 link-local address is FE80::3.

g. Run more tests if it helps visualize the problem. Simulation mode is available.

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Step 3: Propose a solution to solve the problem.

Compare your answers in Step 2 to the documentation you have available for the network.

What is the error?

PC4's IPv6 default gateway is set to FE80::2, which is R2's link-local address and does not exist on PC4's LAN. It should be R3's G0/0 link-local address, FE80::3.

What solution would you propose to correct the problem?

Change PC4's IPv6 default gateway from FE80::2 to FE80::3 (Desktop > IP Configuration > IPv6 Gateway).

Step 4: Implement the plan.

Implement the solution you proposed in Step 3b.

Step 5: Verify that connectivity is restored.

a. From **PC2** test connectivity to **PC4**.

b. From **PC4** test connectivity to **PC2**.

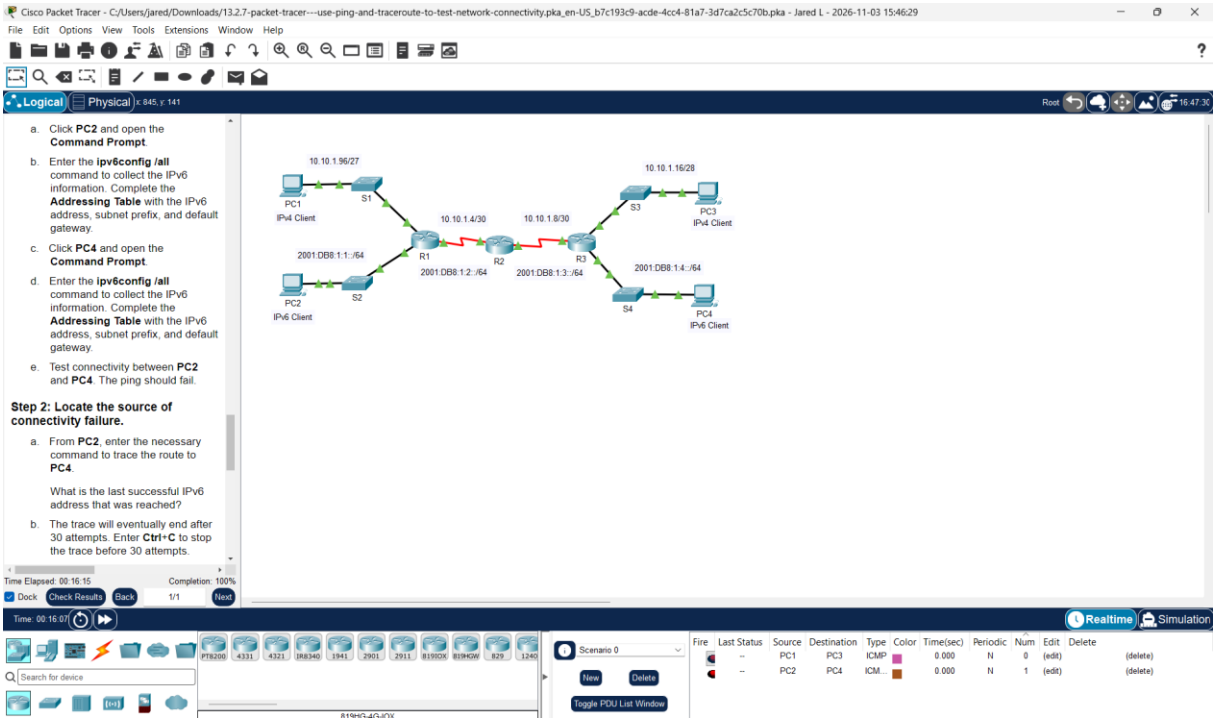
Is the problem resolved?

Yes. Pings from PC2 to PC4 and from PC4 to PC2 now succeed.

Step 6: Document the solution.

IPv6: PC4's default gateway was FE80::2 instead of R3's G0/0 link-local address FE80::3. I changed the gateway on PC4 to FE80::3. PC2 and PC4 can now ping each other.

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